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Processing method, method of manufacturing semiconductor device, processing apparatus, and recording medium

Abstract

There is provide a technique that includes: etching a base on a surface of a substrate by performing a cycle, the cycle including: (a) forming a layer on a surface of the base by exposing the base to a modifying agent; and (b) causing a reaction between a halogen-containing radical and the base by exposing the layer to a halogen-containing gas such that the layer reacts with the halogen-containing gas to generate the halogen-containing radical.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a Continuation application of U.S. patent application Ser. No. 17/693,022 filed Mar. 11, 2022, which is a continuation application of U.S. patent application Ser. No. 16/953,471 filed on Nov. 20, 2020, which is now U.S. Pat. No. 11,315,800 issued on Apr. 26, 2022, which is based upon and claims the benefit of priority from Japanese Patent Application No. 2019-211137, filed on Nov. 22, 2019, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a method of manufacturing a semiconductor device, a substrate processing apparatus, and a recording medium.

BACKGROUND

(2) In the related art, as a process of manufacturing a semiconductor device, a process of etching a base exposed on a surface of a substrate is carried out.

SUMMARY

(3) Some embodiments of the present disclosure provide a technique capable of enhancing etching controllability.

(4) According to an embodiment of the present disclosure, there is provided a technique that includes: etching a base exposed on a surface of a substrate by performing a cycle a predetermined number of times, the cycle including non-simultaneously performing: (a) forming a modified layer on a surface of the base by supplying a modifying agent to the base; and (b) causing a reaction between a halogen-containing radical and the base by supplying a halogen-containing gas to the modified layer such that the modified layer reacts with the halogen-containing gas to generate the halogen-containing radical.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure.

(2) FIG. 1 is a schematic configuration diagram of a vertical type process furnace of a substrate processing apparatus suitably used in an embodiment of the present disclosure, in which a portion of the process furnace 202 is shown in a vertical cross-sectional view.

(3) FIG. 2 is a schematic configuration diagram of a vertical type process furnace of the substrate processing apparatus suitably used in an embodiment of the present disclosure, in which a portion of the process furnace 202 is shown in a cross-sectional view taken along line A-A in FIG. 1.

(4) FIG. 3 is a schematic configuration diagram of a controller 121 of the substrate processing apparatus suitably used in an embodiment of the present disclosure, in which a control system of the controller 121 is shown in a block diagram.

(5) FIG. 4 is a diagram illustrating a gas supply sequence according to an embodiment of the

present disclosure.

(6) FIG. 5A is a partial enlarged cross-sectional view of a surface of a wafer **200** illustrating a state in which a surface of a base **200a** is OH-terminated by pre-processing the surface of the base **200a** exposed on the surface of the wafer **200**, FIG. 5B is a partial enlarged cross-sectional view of the surface of the wafer **200** illustrating a state in which a modified layer **200b** is formed on the surface of the base **200a** by supplying a modifying agent to the base **200a** having a surface OH-terminated, FIG. 5C is a partial enlarged cross-sectional view of the surface of the wafer **200** illustrating a state in which a halogen-containing radical is generated by supplying a halogen-containing gas to the modified layer **200b** such that the modified layer **200b** reacts with the halogen-containing gas, and the halogen-containing radical reacts with the base **200a**, and FIG. 5D is a partial enlarged cross-sectional view of the surface of the wafer **200** after a surface layer of the base **200a** is etched by the reaction with the halogen-containing radical.

(7) FIG. 6 is a view illustrating measurement results of an etching rate of the base.

DETAILED DESCRIPTION

(8) Reference will now be made in detail to various embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure.

However, it will be apparent to one of ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, systems, and components have not been described in detail so as not to unnecessarily obscure aspects of the various embodiments.

An Embodiment of the Present Disclosure

(9) An embodiment of the present disclosure will now be described mainly with reference to FIGS. 1 to 4.

(1) Configuration of the Substrate Processing Apparatus

(10) As illustrated in FIG. 1, a process furnace **202** includes a heater **207** as a heating mechanism (temperature adjustment part). The heater **207** has a cylindrical shape and is supported by a holding plate to be vertically installed. The heater **207** functions as an activation mechanism (an excitation part) configured to thermally activate (excite) a gas.

(11) A reaction tube **203** is disposed inside the heater **207** to be concentric with the heater **207**. The reaction tube **203** is made of a heat resistant material, for example, quartz (SiO₂), silicon carbide (SiC) or the like, and has a cylindrical shape with its upper end closed and its lower end opened. A manifold **209** is disposed below the reaction tube **203** in a concentric relationship with the reaction tube **203**. The manifold **209** is made of a metal material, for example, stainless steel (SUS), and has a cylindrical shape with its upper and lower ends opened. The upper end of the manifold **209** engages with the lower end of the reaction tube **203**. The manifold **209** is configured to support the reaction tube **203**. An O-ring **220a** as a seal member is installed between the manifold **209** and the reaction tube **203**. Similar to the heater **207**, the reaction tube **203** is vertically installed. A processing vessel (reaction vessel) mainly includes the reaction tube **203** and the manifold **209**. A process chamber **201** is formed in a hollow cylindrical portion of the processing vessel. The process chamber **201** is configured to be capable of accommodating wafers **200** as substrates. The processing of the wafers **200** is performed in the process chamber **201**.

(12) Nozzles **249a** to **249c** as first to third supply parts are installed in the process chamber **201** to penetrate a sidewall of the manifold **209**. The nozzles **249a** to **249c** will be referred to as first to third nozzles, respectively. The nozzles **249a** to **249c** are each made of a heat resistant material such as quartz, SiC or the like. Gas supply pipes **232a** to **232c** are connected to the nozzles **249a** to **249c** respectively. The nozzles **249a** to **249c** are different nozzles, in which each of the nozzles **249a** to **249c** is installed adjacent to the nozzle **249b**.

(13) Mass flow controllers (MFCs) **241a** to **241c**, which are flow rate controllers (flow rate control parts), and valves **243a** to **243c**, which are opening/closing valves, are installed at the gas supply

pipes **232a** to **232c** sequentially from the corresponding upstream sides of gas flow, respectively. Gas supply pipes **232d** to **232f** are connected to the gas supply pipes **232a** to **232c** at the downstream side of the valves **243a** to **243c** respectively. MFCs **241d** to **241f** and valves **243d** to **243f** are installed at the gas supply pipes **232d** to **232f** sequentially from the corresponding upstream sides of gas flow, respectively. The gas supply pipes **232a** to **232f** are made of a metal material such as stainless steel (SUS) or the like.

(14) As illustrated in FIG. 2, the nozzles **249a** to **249c** are disposed in a space with an annular plan-view shape between the inner wall of the reaction tube **203** and the wafers **200** such that the nozzles **249a** to **249c** extend upward along an arrangement direction of the wafers **200** from a lower portion to an upper portion of the inner wall of the reaction tube **203**. Specifically, the nozzles **249a** to **249c** are installed at a lateral side of a wafer arrangement region in which the wafers **200** are arranged, namely in a region which horizontally surrounds the wafer arrangement region, to extend along the wafer arrangement region. The nozzle **249b** is disposed to face an exhaust port **231a**, which will be described later, on a straight line in a plane view, with centers of the wafers **200** loaded into the process chamber **201** interposed therebetween. The nozzles **249a** and **249c** are disposed to sandwich a straight line L passing through the nozzle **249b** and a center of the exhaust port **231a** from both sides along the inner wall of the reaction tube **203** (an outer peripheral portion of the wafers **200**). The straight line L is also a straight line passing through the nozzle **249b** and the centers of the wafers **200**. That is, it may be said that the nozzle **249c** is installed at the opposite side of the nozzle **249a** with the straight line L interposed therebetween. The nozzles **249a** and **249c** are disposed in line symmetry with the straight line L as a symmetry axis. Gas supply holes **250a** to **250c** configured to supply a gas are installed on the side surfaces of the nozzles **249a** to **249c**, respectively. The gas supply holes **250a** to **250c** are opened to face the exhaust port **231a** in the plane view, thus allowing a gas to be supplied toward the wafers **200**. The gas supply holes **250a** to **250c** may be formed in a plural number between the lower portion and the upper portion of the reaction tube **203**.

(15) A silicon (Si)-containing gas as a modifying gas is supplied from the gas supply pipe **232a** into the process chamber **201** via the MFC **241a**, the valve **243a**, and the nozzle **249a**. As the Si-containing gas, an aminosilane-based gas which is a gas containing Si and an amino group may be used. The amino group refers to a functional group obtained by coordinating one or two hydrocarbon groups containing one or more C atoms to one N atom (a functional group obtained by substituting one or both of H of an amino group represented by NH_2 by a hydrocarbon group containing one or more C atoms). When two hydrocarbon groups constituting a portion of the amino group are coordinated to one N, the two may be the same hydrocarbon group or different hydrocarbon groups. The hydrocarbon group may contain a single bond like an alkyl group, or may contain an unsaturated bond such as a double bond or a triple bond. The amino group may have a cyclic structure. Since the amino group is bonded to Si, which is a central atom of an aminosilane molecule, the amino group in aminosilane may be referred to as a ligand or an amino ligand. The aminosilane-based gas may further contain a hydrocarbon group, in addition to containing Si and an amino group. The hydrocarbon group may contain a single bond like an alkyl group, or may contain an unsaturated bond such as a double bond or a triple bond. The hydrocarbon group may have a cyclic structure. The hydrocarbon group may be bonded to Si, which is the central atom of the aminosilane molecule, and in this case, the hydrocarbon group in aminosilane may be referred to as a ligand or a hydrocarbon ligand. As the aminosilane-based gas, it may be possible to use, for example, a dimethylaminotrimethylsilane ($(\text{CH}_3)_3\text{Si}(\text{CH}_3)_2\text{NH}_2$, abbreviation: DMATMS) gas. In addition to one amino group (dimethylamino group) being bonded to Si, which is a central atom of DMATMS, three alkyl groups (methyl groups) are bonded to Si. That is, DMATMS contains one amino ligand and three alkyl ligands.

(16) A halogen-containing gas is supplied from the gas supply pipe **232b** into the process chamber **201** via the MFC **241b**, the valve **243b**, and the nozzle **249b**. Halogen includes chlorine (Cl),

fluorine (F), iodine (I), and the like. As the halogen-containing gas, it may be possible to use, for example, a fluorine (F.sub.2) gas.

(17) A gas containing oxygen (O) and hydrogen (H) is supplied from the gas supply pipe **232c** into the process chamber **201** via the MFC **241c**, the valve **243c**, and the nozzle **249c**. As the gas containing O and H, it may be possible to use, for example, water vapor (H.sub.2O gas).

(18) An inert gas, for example, a nitrogen (N.sub.2) gas, is supplied from the gas supply pipes **232d** to **232f** into the process chamber **201** via the MFCs **241d** to **241f**, the valves **243d** to **243f**, the gas supply pipes **232a** to **232c**, and the nozzles **249a** to **249c** respectively. The N.sub.2 gas acts as a purge gas, a carrier gas, a dilution gas, or the like.

(19) A modifying agent supply system mainly includes the gas supply pipe **232a**, the MFC **241a**, and the valve **243a**. A halogen-containing gas supply system mainly includes the gas supply pipe **232b**, the MFC **241b**, and the valve **243b**. An O- and H-containing gas supply system mainly includes the gas supply pipe **232c**, the MFC **241c**, and the valve **243c**. An inert gas supply system mainly includes the gas supply pipes **232d** to **232f**, the MFCs **241d** to **241f**, and the valves **243d** to **243f**.

(20) One or all of various supply systems described above may be configured as an integrated supply system **248** in which the valves **243a** to **243f**, the MFCs **241a** to **241f**, and the like are integrated. The integrated supply system **248** is connected to each of the gas supply pipes **232a** to **232f** so that a supply operation of various kinds of gases into the gas supply pipes **232a** to **232f**, i.e., an opening/closing operation of the valves **243a** to **243f**, a flow rate adjusting operation by the MFCs **241a** to **241f** or the like, is controlled by a controller **121** which will be described later. The integrated supply system **248** is configured as an integral type or division type integrated unit, and may be attached to or detached from the gas supply pipes **232a** to **232f** or the like on an integrated unit basis, such that maintenance, replacement, expansion or the like of the integrated supply system **248** may be performed on the integrated unit basis.

(21) The exhaust port **231a** configured to exhaust an internal atmosphere of the process chamber **201** is installed at a lower side of the sidewall of the reaction tube **203**. As illustrated in FIG. 2, the exhaust port **231a** is installed at a position facing the nozzles **249a** to **249c** (the gas supply holes **250a** to **250c**) in the plane view, with the wafers **200** interposed therebetween. The exhaust port **231a** may be installed between the lower portion and the upper portion of the sidewall of the reaction tube **203**, that is, along the wafer arrangement region. An exhaust pipe **231** is connected to the exhaust port **231a**.

(22) A vacuum pump **246** as a vacuum exhaust device is connected to the exhaust pipe **231** via a pressure sensor **245** as a pressure detector (pressure detection part) which detects the internal pressure of the process chamber **201** and an auto pressure controller (APC) valve **244** as a pressure regulator (pressure regulation part). The APC valve **244** is configured so that a vacuum exhaust and a vacuum exhaust stop of the interior of the process chamber **201** can be performed by opening and closing the APC valve **244** while operating the vacuum pump **246** and so that the internal pressure of the process chamber **201** can be adjusted by adjusting an opening degree of the APC valve **244** based on pressure information detected by the pressure sensor **245** while operating the vacuum pump **246**. An exhaust system mainly includes the exhaust pipe **231**, the APC valve **244** and the pressure sensor **245**. The vacuum pump **246** may be regarded as being included in the exhaust system.

(23) A seal cap **219**, which serves as a furnace opening lid configured to be capable of hermetically sealing a lower end opening of the manifold **209**, is installed under the manifold **209**. The seal cap **219** is made of a metal material such as stainless steel (SUS) or the like, and is formed in a disc shape. An O-ring **220b**, which is a seal member making contact with the lower end portion of the manifold **209**, is installed on an upper surface of the seal cap **219**.

(24) A rotation mechanism **267** configured to rotate a boat **217**, which will be described later, is installed under the seal cap **219**. A rotary shaft **255** of the rotation mechanism **267**, which

penetrates the seal cap **219**, is connected to the boat **217**. The rotation mechanism **267** is configured to rotate the wafers **200** by rotating the boat **217**. The seal cap **219** is configured to be vertically moved up or down by a boat elevator **115** which is an elevator mechanism installed outside the reaction tube **203**. The boat elevator **215** is configured as a transfer device (transfer mechanism) which loads and unloads (transfers) the wafers **200** into and from (out of) the process chamber **201** by moving the seal cap **219** up or down.

(25) A shutter **219s** as a furnace opening lid capable of hermetically sealing the lower end opening of the manifold **209**, with the boat **217** unloaded from the interior of the process chamber **201** by moving the seal cap **219** down, is installed under the manifold **209**. The shutter **219s** is made of a metal material such as stainless steel or the like, and is formed in a disc shape. An O-ring **220c** as a seal member making contact with the lower end portion of the manifold **209** is installed on an upper surface of the shutter **219s**. An opening/closing operation (an up-down movement operation or a rotational movement operation) of the shutter **219s** is controlled by a shutter opening/closing mechanism **115s**.

(26) The boat **217** serving as a substrate support is configured to support a plurality of wafers **200**, for example, 25 to 200 wafers, in such a state that the wafers **200** are arranged in a horizontal posture and in multiple stages along a vertical direction with the centers of the wafers **200** aligned with one another. That is, the boat **217** is configured to arrange the wafers **200** to be spaced apart from each other. The boat **217** is made of a heat resistant material such as quartz or SiC. Heat insulating plates **218** made of a heat resistant material such as quartz or SiC are installed below the boat **217** in multiple stages.

(27) A temperature sensor **263** serving as a temperature detector is installed in the reaction tube **203**. Based on temperature information detected by the temperature sensor **263**, a degree of supplying electric power to the heater **207** is adjusted such that the interior of the process chamber **201** has a desired temperature distribution. The temperature sensor **263** is installed along the inner wall of the reaction tube **203**.

(28) As illustrated in FIG. 3, the controller **121**, which is a control part (control means), may be configured as a computer including a central processing unit (CPU) **121a**, a random access memory (RAM) **121b**, a memory device **121c**, and an I/O port **121d**. The RAM **121b**, the memory device **121c** and the I/O port **121d** are configured to be capable of exchanging data with the CPU **121a** via an internal bus **121e**. An input/output device **122** configured as, for example, a touch panel or the like, is connected to the controller **121**.

(29) The memory device **121c** includes, for example, a flash memory, a hard disk drive (HDD), or the like. A control program that controls operations of a substrate processing apparatus, a process recipe in which sequences, conditions and the like of substrate processing to be described below, and the like are readably stored in the memory device **121c**. The process recipe functions as a program configured to be capable of causing the controller **121** to execute each sequence in the substrate processing, which will be described below, to obtain a predetermined result. Hereinafter, the process recipe and the control program will be generally and simply referred to as a “program.” Furthermore, the process recipe will be simply referred to as a “recipe.” When the term “program” is used herein, it may indicate a case of including only the recipe, a case of including only the control program, or a case of including both the recipe and the control program. The RAM **121b** is configured as a memory area (work area) in which a program, data and the like read by the CPU **121a** is temporarily stored.

(30) The I/O port **121d** is connected to the MFCs **241a** to **241f**, the valves **243a** to **243f**, the pressure sensor **245**, the APC valve **244**, the vacuum pump **246**, the temperature sensor **263**, the heater **207**, the rotation mechanism **267**, the boat elevator **115**, the shutter opening/closing mechanism **115s**, and the like, as described above.

(31) The CPU **121a** is configured to read the control program from the memory device **121c** and execute the same. The CPU **121a** also reads the recipe from the memory device **121c** according to

an input of an operation command from the input/output device **122**. In addition, the CPU **121a** is configured to control, according to the contents of the recipe thus read, the flow rate adjusting operation of various kinds of gases by the MFCs **241a** to **241f**, the opening/closing operation of the valves **243a** to **243f**, the opening/closing operation of the APC valve **244**, the pressure regulating operation performed by the APC valve **244** based on the pressure sensor **245**, the driving and stopping of the vacuum pump **246**, the temperature adjusting operation performed by the heater **207** based on the temperature sensor **263**, the operation of rotating the boat **217** and adjusting the rotation speed of the boat **217** with the rotation mechanism **267**, the operation of moving the boat **217** up or down with the boat elevator **115**, the operation of opening and closing the shutter **219s** with the shutter opening/closing mechanism **115s**, and the like.

(32) The controller **121** may be configured by installing, on the computer, the aforementioned program stored in an external memory device **123**. The external memory device **123** may include, for example, a magnetic disc such as an HDD, an optical disc such as a CD, a magneto-optical disc such as an MO, a semiconductor memory such as a USB memory, and the like. The memory device **121c** or the external memory device **123** is configured as a computer-readable recording medium. Hereinafter, the memory device **121c** and the external memory device **123** will be generally and simply referred to as a “recording medium.” When the term “recording medium” is used herein, it may indicate a case of including only the memory device **121c**, a case of including only the external memory device **123**, or a case of including both the memory device **121c** and the external memory device **123**. Furthermore, the program may be supplied to the computer by using a communication means such as the Internet or a dedicated line, instead of using the external memory device **123**.

(2) Substrate Processing

(33) A processing sequence example of etching a base **200a** exposed on a surface of a wafer **200** as a substrate by using the aforementioned substrate processing apparatus, which is a process for manufacturing a semiconductor device, will be described mainly with reference to FIGS. **4** and **5A** to **5D**. In the following descriptions, the operations of the respective parts constituting the substrate processing apparatus are controlled by the controller **121**.

(34) In the processing sequence illustrated in FIG. **4**, a cycle, which includes non-simultaneously performing: a step of forming a modified layer **200b** on a surface of the base **200a** by supplying a DMATMS gas as a modifying agent to a base **200a** exposed on a surface of a wafer **200** (DMATMS gas supply); and a step of causing a reaction between a F-containing radical (F*) as a halogen-containing radical and the base **200a** by supplying a F.sub.2 gas as a halogen-containing gas to the modified layer **200b** such that the modified layer **200b** reacts with the F.sub.2 gas to generate the F-containing radical (F*) (F.sub.2 gas supply), is performed a predetermined number of times (n times, where n is an integer of 1 or more).

(35) Further, the cycle of the processing sequence illustrated in FIG. **4** performs a step of pre-processing the surface of the base **200a** by supplying an H.sub.2O gas as a gas containing O and H to the base **200a** exposed on the surface of the wafer **200** before performing the DMATMS gas supply (H.sub.2O gas supply).

(36) Further, in the processing sequence illustrated in FIG. **4**, each step is performed in a non-plasma atmosphere.

(37) In the present disclosure, for the sake of convenience, the processing sequence described above may be denoted as follows. The same denotation will be used in the modifications and the like as described below.

(H.sub.2O.fwdarw.DMATMS.fwdarw.F.sub.2)×n

(38) When the term “wafer” is used herein, it may refer to a wafer itself or a laminated body of a wafer and a layer or film formed on the surface of the wafer. In addition, when the phrase “a surface of a wafer” is used herein, it may refer to a surface of a wafer itself or a surface of a layer or the like formed on a wafer. Further, in the present disclosure, the expression “a layer is formed

on a wafer” may mean that a layer is directly formed on a surface of a wafer itself or that a layer is formed on a layer or the like formed on a wafer. In addition, when the term “substrate” is used herein, it may be synonymous with the term “wafer.”

(39) Further, when the term “base” is used herein, it may refer to a wafer itself or a layer or film formed on the surface of the wafer. In addition, when the phrase “a surface of a base” is used herein, it may refer to a surface of a wafer itself or a surface of a layer or the like formed on the surface of the wafer. Furthermore, in the present disclosure, the expression “a layer is formed on a surface of a base” may mean that a layer is directly formed on a surface of a wafer itself or that a layer is formed on a surface of a layer or the like formed on the surface of the wafer.

(40) (Wafer Charging and Boat Loading)

(41) When a plurality of wafers **200** is charged on the boat **217** (wafer charging), the shutter **219s** may be moved by the shutter opening/closing mechanism **115s** to open the lower end opening of the manifold **209** (shutter opening). Thereafter, as illustrated in FIG. 1, the boat **217** supporting the plurality of wafers **200** is lifted up by the boat elevator **115** and is loaded into the process chamber **201** (boat loading). In this state, the seal cap **219** seals the lower end of the manifold **209** via the O-ring **220b**. Furthermore, for example, a silicon oxide film (SiO film) is exposed as the base **200a** to be etched on the surface of each of the wafers **200**.

(42) (Pressure Regulation and Temperature Adjustment)

(43) The interior of the process chamber **201**, which is a space where the wafers **200** are placed, is vacuum-exhausted (depressurization-exhausted) by the vacuum pump **246** to reach a desired pressure (degree of vacuum). In this operation, the internal pressure of the process chamber **201** is measured by the pressure sensor **245**. The APC valve **244** is feedback-controlled based on the measured pressure information. Further, the wafers **200** in the process chamber **201** are heated by the heater **207** to have a desired processing temperature. In this operation, the degree of supplying electric power to the heater **207** is feedback-controlled based on the temperature information detected by the temperature sensor **263** so that the interior of the process chamber **201** has a desired temperature distribution. Further, the rotation of the wafers **200** by the rotation mechanism **267** begins. The exhaust of the interior of the process chamber **201** and the heating and rotation of the wafers **200** may be all continuously performed at least until the processing of the wafers **200** is completed.

(44) (Etching)

(45) Next, the following steps 1 to 3 are sequentially performed.

(46) [Step 1]

(47) At this step, an H.sub.2O gas is supplied to the wafer **200** in the process chamber **201**, that is, to the base **200a** exposed on the surface of the wafer **200**.

(48) Specifically, the valve **243c** is opened to allow an H.sub.2O gas to flow through the gas supply pipe **232c**. A flow rate of the H.sub.2O gas is adjusted by the MFC **241c**. The H.sub.2O gas is supplied into the process chamber **201** via the nozzle **249c** and is exhausted from the exhaust port **231a**. At this time, the H.sub.2O gas is supplied to the wafer **200** (H.sub.2O gas supply).

Simultaneously, the valves **243d** to **243f** may be opened to supply an N.sub.2 gas into the process chamber **201** via the nozzles **249a** to **249c** respectively.

(49) By supplying the H.sub.2O gas to the wafer **200** under a condition to be described below, the surface of the base **200a** exposed on the surface of the wafer **200** can be pre-processed.

Specifically, as illustrated in FIG. 5A, the surface of the base **200a** can be terminated with a hydroxy group (OH group), that is, OH-terminated. Further, the OH termination formed on the surface of the base **200a** functions as an adsorption site that promotes adsorption of Si contained in the DMATMS gas on the surface of the base **200a** in the DMATMS gas supply to be described below.

(50) After the surface of the base **200a** is pre-processed, the valve **243c** is closed to stop the supply of the H.sub.2O gas into the process chamber **201**. Then, the interior of the process chamber **201** is

vacuum-exhausted and the gas or the like remaining within the process chamber **201** is removed from the interior of the process chamber **201**. At this time, the valves **243d** to **243f** are opened to supply an N.sub.2 gas into the process chamber **201** via the nozzles **249a** to **249c**. The N.sub.2 gas supplied from the nozzles **249a** to **249c** acts as a purge gas. Thus, the interior of the process chamber **201** is purged (purge).

(51) The processing condition in the H.sub.2O gas supply may be exemplified as follows:

H.sub.2O gas supply flow rate: 10 to 2,000 sccm Supply time of H.sub.2O gas: 5 to 1,800 seconds

N.sub.2 gas supply flow rate (per gas supply pipe): 0 to 10,000 sccm Processing temperature: 30 to 300 degrees C. Processing pressure: 5 to 1,000 Pa.

(52) Furthermore, in the present disclosure, an expression of a numerical range such as “5 to 1,000 Pa” may mean that a lower limit value and an upper limit value are included in that range.

Therefore, for example, “5 to 1,000 Pa” may mean “5 Pa or more and 1,000 Pa or less”. The same applies to other numerical ranges.

(53) As the gas containing O and H, it may be possible to use, in addition to the H.sub.2O gas, for example, an O-containing gas containing an O—H bond such as a hydrogen peroxide

(H.sub.2O.sub.2) gas or the like, that is, a gas containing an OH group. Further, as the gas containing O and H, it may be possible to use an O-containing gas and an H-containing gas such as an O.sub.2 gas+H.sub.2 gas.

(54) As the inert gas, in addition to the N.sub.2 gas, a rare gas such as an Ar gas, a He gas, a Ne gas, a Xe gas or the like may be used. This also applies to each step to be described below.

(55) [Step 2]

(56) After step 1 is completed, a DMATMS gas is supplied to the wafer **200** in the process chamber **201**, that is, to the base **200a** whose surface is OH-terminated.

(57) Specifically, the valve **243a** is opened to allow a DMATMS gas to flow through the gas supply pipe **232a**. A flow rate of the DMATMS gas is adjusted by the MFC **241a**. The DMATMS gas is supplied into the process chamber **201** via the nozzle **249a** and is exhausted from the exhaust port **231a**. At this time, the DMATMS gas is supplied to the wafer **200** (DMATMS gas supply). At this time, the valves **243d** to **243f** may be opened to supply an N.sub.2 gas into the process chamber **201** via the nozzles **249a** to **249c** respectively.

(58) By supplying the DMATMS gas to the wafer **200** under a condition to be described below, as illustrated in FIG. 5B, the surface of the base **200a** can be modified to allow Si contained in the DMATMS gas, that is, Si contained in aminosilane, to be adsorbed on the surface of the base **200a**. Thus, a modified layer **200b** can be formed on the surface of the base **200a**. Further, the modification referred to herein may mean that the DMATMS gas is physisorbed or chemisorbed on atoms of the surface layer of the base **200a** or that the DMATMS gas and the atoms on the surface layer of the base **200a** are chemically reacted with each other to form a compound. In addition, Si adsorbed on the surface of the base **200a** acts to promote the generation of a fluorine radical (F*) contributing to the etching of the base **200a** in the F.sub.2 gas supply to be described below.

(59) When the supply of the DMATMS gas is continued for a predetermined time, the adsorption of Si on the surface of the base **200a** is saturated. That is, the adsorption of Si on the surface of the base **200a** is self-limited. In other words, when a Si layer having a predetermined saturation thickness of one atomic layer or less than one atomic layer is formed on the surface of the base **200a**, Si is no longer adsorbed on the surface of the base **200a**. The amount of Si adsorbed on the surface of the base **200a**, that is, the amount of Si contained in the modified layer **200b**, becomes a substantially uniform amount over the entire surface of the base **200a**. This is because the processing condition at this step is set to a condition under which the DMATMS gas is not gas-phase decomposed in the process chamber **201**, as will be described later. At this step, since the DMATMS gas is not gas-phase decomposed in the process chamber **201**, multiple deposition of Si contained in DMATMS on the surface of the base **200a** is suppressed. Further, by supplying the DMATMS gas under such a processing condition, the modified layer **200b** formed on the surface of

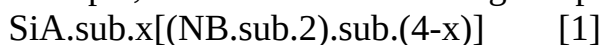
the base **200a** becomes a layer containing a hydrocarbon group such as an alkyl group (a methyl group in this case), in addition to Si contained in the DMATMS gas. Further, the surface of the modified layer **200b** is terminated with a hydrocarbon group or the like. In addition, when Si contained in the DMATMS gas is adsorbed on the base **200a**, the amino group (a dimethylamino group in this case) or the like desorbed from Si is combined with H or the like contained in the OH group terminating the surface of the base **200a** to become a gaseous substance and is exhausted from the exhaust port **231a**.

(60) After the formation of the modified layer **200b** on the surface of the base **200a** is completed, the valve **243a** is closed to stop the supply of the DMATMS gas into the process chamber **201**. Then, the interior of the process chamber **201** is vacuum-exhausted and the gas or the like, which remains within the process chamber **201**, is removed from the interior of the process chamber **201** according to the same processing procedure as that of the purge at step 1 (purge).

(61) The processing condition in the DMATMS gas supply may be exemplified as follows: DMATMS gas supply flow rate: 10 to 2,000 sccm Supply time of DMATMS gas: 5 to 1,800 seconds Processing temperature: 30 to 300 degrees C., specifically 30 to 200 degrees C. in some embodiments Processing pressure: 1 to 5,000 Pa, specifically 5 to 1,000 Pa in some embodiments.

(62) Other processing conditions may be similar to the processing conditions in the H.sub.2O gas supply of step 1. As described above, the condition described herein is the condition under which the DMATMS gas is not gas-phase decomposed in the process chamber **201**, that is, a condition under which the adsorption of Si on the surface of the base **200a** is self-limited and the adsorption of Si is saturated.

(63) As the modifying agent, it may be possible to use, in addition to the DMATMS gas, for example, the aminosilane-based gas represented by the following chemical formula [1].



(64) In the formula [1], A indicates a hydrogen atom, an alkyl group such as a methyl group, an ethyl group, a propyl group or a butyl group, or an alkoxy group such as a methoxy group, an ethoxy group, a propoxy group or a butoxy group. The alkyl group may be not only a linear alkyl group but also a branched alkyl group such as an isopropyl group, an isobutyl group, a secondary butyl group, a tertiary butyl group or the like. The alkoxy group may be not only a linear alkoxy group but also a branched alkoxy group such as an isopropoxy group, an isobutoxy group or the like. B indicates a hydrogen atom or an alkyl group such as a methyl group, an ethyl group, a propyl group, a butyl group or the like. The alkyl group may be not only a linear alkyl group but also a branched alkyl group such as an isopropyl group, an isobutyl group, a secondary butyl group, a tertiary butyl group or the like. A plurality of A's may be identical or different, and two B's may be identical or different. Further, x is an integer of 1 to 3.

(65) [Step 3]

(66) After step 2 is completed, a F.sub.2 gas is supplied to the wafer **200** in the process chamber **201**, namely to the modified layer **200b** formed on the surface of the base **200a**.

(67) Specifically, the valve **243b** is opened to allow the F.sub.2 gas to flow through the gas supply pipe **232b**. A flow rate of the F.sub.2 gas is adjusted by the MFC **241b**. The F.sub.2 gas is supplied into the process chamber **201** via the nozzle **249b** and is exhausted from the exhaust port **231a**. At this time, the F.sub.2 gas is supplied to the wafer **200** (F.sub.2 gas supply). At this time, the valves **243d** to **243f** may be opened to supply a N.sub.2 gas into the process chamber **201** via the nozzles **249a** to **249c** respectively.

(68) By supplying the F.sub.2 gas to the wafer **200** under a condition to be described below, as illustrated in FIG. 5C, the modified layer **200b** may react with the F.sub.2 gas to generate a fluorine-containing radical (F*), thus causing a reaction between the F* and the base **200a**. Then, the surface of the base **200a** may be etched with the F* generated by the reaction between the modified layer **200b** and the F.sub.2 gas. The Si contained in the modified layer **200b** is desorbed from the surface of the base **200a** by the reaction with the F.sub.2 gas and is exhausted from the

exhaust port **231a**, for example, by forming a compound with F.

(69) Further, by etching the base **200a** with F* generated by the reaction between the modified layer **200b** and the F.sub.2 gas, it is possible to enhance a controllability of an etching amount when etching the base **200a**. This is because when all Si contained in the modified layer **200b** is desorbed (consumed) from the base **200a** by the aforementioned reaction, the generation of F* is terminated. Therefore, by controlling the amount of Si contained in the modified layer **200b** to be a predetermined amount as in the present embodiment, the amount of F* generated per this step can be limited to become a corresponding amount, and as a result, the etching amount of the base **200a** with F* per this step can be precisely controlled. As in the present embodiment, when the amount of Si contained in the modified layer **200b** in the DMATMS gas supply is set to a constant amount of one atomic layer or less than one atomic layer over the entire in-plane region of the wafer, the amount of F* generated per this step becomes a constant amount corresponding to the amount over the entire in-plane region of the wafer, and as a result, the etching amount of the base **200a** with F* can be controlled to be a constant amount of, for example, one atomic layer (molecular layer) or less than one atomic layer (molecular layer), over the entire in-plane region of the wafer.

(70) The processing condition in the F.sub.2 gas supply may be exemplified as follows: F.sub.2 gas supply flow rate: 10 to 1,000 sccm Supply time of F.sub.2 gas: 5 to 600 seconds Processing temperature: 30 to 250 degrees C., specifically 30 to 200 degrees C. in some embodiments Processing pressure: 5 to 1,000 Pa.

(71) Other conditions may be similar to the processing conditions in the H.sub.2O gas supply of step 1.

(72) Further, the processing condition at this step is set to a condition under which the reaction between the F.sub.2 gas and the modified layer **200b** proceeds further than the reaction between the F.sub.2 gas and the base **200a**. In other words, the processing condition at this step is set to a condition under which an amount of F.sub.2 gas that reacts with the modified layer **200b** in the reaction between the F.sub.2 gas and the modified layer **200b** is larger than an amount of F.sub.2 gas that reacts with the base **200a** in the reaction between the F.sub.2 gas and the base **200a**. This makes it possible to promote the generation of F* contributing to the etching of the base **200a** while suppressing a direct etching of the base **200a** with the F.sub.2 gas. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a**.

(73) Further, the processing condition at this step can be set to a condition under which the reaction between the F.sub.2 gas and the modified layer **200b** proceeds and the reaction between the F.sub.2 gas and the base **200a** does not proceed. This makes it possible to promote the generation of F* contributing to the etching of the base **200a** while reliably suppressing the direct etching of the base **200a** with the F.sub.2 gas. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a**.

(74) Further, the processing condition at this step is set to a condition under which the reaction between F* and the base **200a** proceeds further than the reaction between the F.sub.2 gas and the base **200a**. In other words, the processing condition at this step is set to a condition under which an amount of F* that reacts with the base **200a** in the reaction between F* and the base **200a** is larger than an amount of F.sub.2 gas that reacts with the base **200a** in the reaction between the F.sub.2 gas and the base **200a**. This makes it possible to suppress the direct etching of the base **200a** with the F.sub.2 gas and to allow the etching reaction of the base **200a** with F* to occur dominantly. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a**.

(75) Further, the processing condition in this step can be set to a condition under which the reaction between F* and the base **200a** proceeds and the reaction between the F.sub.2 gas and the base **200a** does not proceed. This makes it possible to reliably suppress the direct etching of the base **200a** with the F.sub.2 gas and to allow the etching reaction of the base **200a** with F* to occur more dominantly. As a result, it is possible to further enhance the controllability of the etching amount

when etching the base **200a**.

(76) After the etching of the base **200a** with F* generated by the reaction between the modified layer **200b** and the F.sub.2 gas is completed, the valve **243b** is closed to stop the supply of the F.sub.2 gas into the process chamber **201**. Then, the interior of the process chamber **201** is vacuum-exhausted such that the gas or the like, which remains within the process chamber **201**, is removed from the interior of the process chamber **201** according to the same processing procedure as that of the purge at step 1 (purge).

(77) As the halogen-containing gas, it may be possible to use, in addition to the F.sub.2 gas, a chlorine trifluoride (ClF.sub.3) gas, a nitrosyl fluoride (FNO) gas, a chlorine fluoride (ClF) gas, a nitrogen trifluoride (NF.sub.3) gas, a hydrogen fluoride (HF) gas, a chlorine (Cl.sub.2) gas, a hydrogen chloride (HCl) gas, an iodine heptafluoride (IF.sub.7) gas, an iodine pentafluoride (IF.sub.5) gas, or the like. These gases may react with the modified layer **200b** to generate a chlorine-containing radical (Cl*), an iodine-containing radical (I*) or the like, in addition to F*, as the halogen-containing radical, and these radicals may react with the base **200a** to proceed the etching of the base **200a**.

(78) [Performing a Predetermined Number of Times]

(79) A cycle which non-simultaneously, that is, non-synchronously, performs steps 1 to 3 described above is performed a predetermined number of times (n times, where n is an integer of 1 or more). Thus, the base **200a** exposed on the surface of the wafer **200** can be etched at a desired depth, as illustrated in FIG. 5D. The aforementioned cycle may be repeated multiple times. That is, a thickness of a layer etched per one cycle may be set smaller than a desired thickness, and the aforementioned cycle may be repeated multiple times until a thickness of a layer removed by etching becomes equal to the desired thickness.

(80) (After-Purge and Returning to Atmospheric Pressure)

(81) After the etching of the base **200a** is completed, the N.sub.2 gas as a purge gas is supplied from each of the nozzles **249a** to **249c** into the process chamber **201** and is exhausted from the exhaust port **231a**. Thus, the interior of the process chamber **201** is purged and the gas or the reaction byproduct, which remains within the process chamber **201**, is removed from the interior of the process chamber **201** (after-purge). Thereafter, the internal atmosphere of the process chamber **201** is substituted by an inert gas (inert gas substitution). The internal pressure of the process chamber **201** is returned to an atmospheric pressure (returning to atmospheric pressure).

(82) (Boat Unloading and Wafer Discharging)

(83) Thereafter, the seal cap **219** is moved down by the boat elevator **115** to open the lower end of the manifold **209**. Then, the processed wafers **200** supported on the boat **217** are unloaded from the lower end of the manifold **209** to the outside of the reaction tube **203** (boat unloading). After the boat unloading, the shutter **219s** is moved so that the lower end opening of the manifold **209** is sealed by the shutter **219s** via the O-ring **220c** (shutter closing). The processed wafers **200** are unloaded to the outside of the reaction tube **203**, and then are subsequently discharged from the boat **217** (wafer discharging).

(3) Effects According to the Embodiments

(84) According to the embodiments, one or more effects as set forth below may be achieved.

(85) (a) By performing a cycle a predetermined number of times, the cycle including non-simultaneously performing: supplying the DMATMS gas to form the modified layer **200b** on the surface of the base **200a** exposed on the surface of the wafer **200**; and supplying the F.sub.2 gas to cause the reaction between the base **200a** and F* generated by the reaction between the modified layer **200b** and the F.sub.2 gas, it is possible to precisely control the etching amount of the base **200a**. That is, by etching the base **200a** with F* generated by the reaction between the modified layer **200b** and the F.sub.2 gas, it is possible to control the etching amount of the base **200a** at a level of, for example, one atomic layer (molecular layer) or less.

(86) (b) By supplying the F.sub.2 gas under the condition in which the reaction between the F.sub.2

gas and the modified layer **200b** proceeds further than the reaction between the F.sub.2 gas and the base **200a** (that is, the condition in which an amount of F.sub.2 gas that reacts with the modified layer **200b** in the reaction between the F.sub.2 gas and the modified layer **200b** is larger than an amount of F.sub.2 gas that reacts with the base **200a** in the reaction between the F.sub.2 gas and the base **200a**), it is possible to suppress the etching amount of the base **200a** from fluctuating depending on an exposure amount of the F.sub.2 gas. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a**. For example, by supplying the F.sub.2 gas under the aforementioned processing condition even under a situation where a 3D structure such as a trench or a hole is formed on the surface of the base **200a** and the exposure amount of the F.sub.2 gas is likely to be locally reduced or under a situation where a partial pressure of the F.sub.2 gas in the process chamber **201** is not constant depending on a position, it is possible to control the etching amount of the base **200a** to be uniform over the in-plane of the wafer.

(87) (c) By supplying the F.sub.2 gas under the condition in which the reaction between the F.sub.2 gas and the modified layer **200b** proceeds and the reaction between the F.sub.2 gas and the base **200a** does not proceed, it is possible to reliably suppress the etching amount of the base **200a** from fluctuating depending on an exposure amount of the F.sub.2 gas. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a**. For example, by supplying the F.sub.2 gas under the aforementioned processing condition even under a situation where a 3D structure such as a trench or a hole is formed on the surface of the base **200a** and the exposure amount of the F.sub.2 gas is likely to be locally reduced or under a situation where the partial pressure of the F.sub.2 gas in the process chamber **201** is not constant depending on a position, it is possible to reliably control the etching amount of the base **200a** to be uniform over the in-plane of the wafer.

(88) (d) By supplying the F.sub.2 gas supply under the condition in which the reaction between F* and the base **200a** proceeds further than the reaction between the F.sub.2 gas and the base **200a** (that is, the condition in which an amount of F* that reacts with the base **200a** in the reaction between F* and the base **200a** is larger than an amount of F.sub.2 gas that reacts with the base **200a** in the reaction between the F.sub.2 gas and the base **200a**), it is possible to suppress the etching amount of the base **200a** from fluctuating depending on an exposure amount of the F.sub.2 gas. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a**. For example, by supplying the F.sub.2 gas under the aforementioned processing condition even under a situation where a 3D structure such as a trench or a hole is formed on the surface of the base **200a** and the exposure amount of the F.sub.2 gas is likely to be locally reduced or under a situation where the partial pressure of the F.sub.2 gas in the process chamber **201** is not constant depending on a position, it is possible to control the etching amount of the base **200a** to be uniform over the in-plane of the wafer.

(89) (e) By supplying the F.sub.2 gas under the condition in which the reaction between F* and the base **200a** proceeds and the reaction between the F.sub.2 gas and the base **200a** does not proceed, it is possible to reliably suppress the etching amount of the base **200a** from fluctuating depending on an exposure amount of the F.sub.2 gas. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a**. For example, by supplying the F.sub.2 gas under the aforementioned processing condition even under a situation where a 3D structure such as a trench or a hole is formed on the surface of the base **200a** and the exposure amount of the F.sub.2 gas is likely to be locally reduced or under a situation where the partial pressure of the F.sub.2 gas in the process chamber **201** is not constant depending on a position, it is possible to reliably control the etching amount of the base **200a** to be uniform over the in-plane of the wafer.

(90) (f) In the DMATMS gas supply, Si contained in the DMATMS gas is adsorbed on the surface of the base **200a** by using the DMATMS gas, which is the Si-containing gas, as the modifying

agent, whereby it is possible to promote the generation of F* contributing to the etching of the base **200a** in the subsequent F.sub.2 gas supply. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a**.

(91) (g) In the DMATMS gas supply, Si contained in the DMATMS gas is adsorbed on the surface of the base **200a** by using the DMATMS gas, which is the aminosilane-based gas, as the modifying agent, whereby it is possible to improve the uniformity in the thickness of the modified layer **200b** formed on the surface of the base **200a** in the plane of the wafer, that is, the uniformity of the adsorption amount of Si in the plane of the wafer. This makes it possible to make the amount of F* generated in the F.sub.2 gas supply uniform in the plane of the wafer. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a** over the in-plane of the wafer, and to reliably control the etching amount of the base **200a** at a level of, for example, one atomic layer (molecular layer) or less, over the in-plane of the wafer.

(92) (h) In the DMATMS gas supply, the DMATMS gas is supplied under the condition in which the adsorption of Si on the surface of the base **200a** is saturated, that is, the condition in which the adsorption of Si is self-limited, whereby it is possible to further improve uniformity of the thickness of the modified layer **200b** formed on the surface of the base **200a** in the plane of the wafer, that is, the uniformity of the adsorption amount of Si in the plane of the wafer. Thus, it is possible to make the amount of F* generated in the F.sub.2 gas supply uniform in the plane of the wafer. As a result, it is possible to further enhance the controllability of the etching amount when etching the base **200a** over the in-plane of the wafer, and to reliably control the etching amount of the base **200a** at a level of, for example, one atomic layer (molecular layer) or less, over the in-plane of the wafer.

(93) (i) By performing the H.sub.2O gas supply which pre-processes the surface of the base **200a** and OH-terminates the same before performing the DMATMS gas supply, it is possible to promote the formation of the modified layer **200b** on the surface of the base **200a** in the subsequent DMATMS gas supply.

(94) (j) By performing each step in a non-plasma atmosphere, it is possible to enhance the controllability of the processing at each step, and as a result, to further enhance the controllability of the etching amount. Further, it is also possible to avoid plasma damage to the wafer **200** or to each base exposed on the surface of the wafer **200**.

(95) (k) The effects mentioned above can be similarly achieved in the case where a modifying agent other than the DMATMS gas is used, in the case where a halogen-containing gas other than the F.sub.2 gas is used, in the case where a gas containing O and H other than the H.sub.2O gas is used, or in the case where an inert gas other than the N.sub.2 gas is used.

Other Embodiments of the Present Disclosure

(96) While embodiments of the present disclosure have been specifically described above, the present disclosure is not limited to the aforementioned embodiments but may be variously modified without departing from the spirit of the present disclosure.

(97) For example, the H.sub.2O gas supply may not be performed in each cycle as in a substrate processing sequence described below. Even in this case, the same effects as those of the aforementioned embodiments may be achieved. Further, it is possible to shorten a cycle time and to improve a productivity of substrate processing. However, in the DMATMS gas supply, the H.sub.2O gas supply may be performed before performing the DMATMS gas supply, as in the aforementioned embodiments to promote the formation of the modified layer **200b** on the surface of the base **200a** exposed on the surface of the wafer **200**.

(DMATMS.fwdarw.F.sub.2) $\times$ n

(98) Further, for example, the base **200a** exposed on the surface of the wafer **200** is not limited to the SiO film but may be a film containing a semi-metal element such as a silicon oxycarbonitride film (SiOCN film), a silicon oxynitride film (SiON film), a silicon oxycarbide film (SiOC film), a silicon nitride film (SiN film), a silicon carbide film (SiC film), a silicon carbonitride film (SiCN

film), a silicon boronitride film (SiBN film), a silicon borocarbonitride film (SiBCN film), a germanium film (Ge film) or a silicon germanium film (SiGe film), or a film containing a metal element such as a titanium nitride film (TiN film) or a tungsten film (W film). Even in these cases, the same effects as those of the aforementioned embodiments may be achieved.

(99) Recipes used in each processing may be provided individually according to the processing contents and may be stored in the memory device **121c** via a telecommunication line or the external memory device **123**. Moreover, at the start of each processing, the CPU **121a** may properly select an appropriate recipe from the recipes stored in the memory device **121c** according to the processing contents. Thus, it is possible for a single substrate processing apparatus to realize various etching processes with enhanced reproducibility. In addition, it is possible to reduce an operator's burden and to quickly start each processing while avoiding an operation error.

(100) The recipes mentioned above are not limited to newly-prepared ones but may be provided by, for example, modifying the existing recipes already installed in the substrate processing apparatus. When modifying the recipes, the modified recipes may be installed in the substrate processing apparatus via a telecommunication line or a recording medium storing the recipes. In addition, the existing recipes already installed in the substrate processing apparatus may be directly modified by operating the input/output device **122** of the existing substrate processing apparatus.

(101) In the aforementioned embodiments, there has been described an example in which the etching process is performed by using a batch-type substrate processing apparatus capable of processing a plurality of substrates at a time. The present disclosure is not limited to the aforementioned embodiments but may be appropriately applied to, for example, a case where the etching process is performed by using a single-wafer-type substrate processing apparatus capable of processing a single substrate or several substrates at a time. In addition, in the aforementioned embodiments, there has been described an example in which the etching process is performed by using the substrate processing apparatus provided with a hot-wall-type process furnace. The present disclosure is not limited to the aforementioned embodiments but may be appropriately applied to a case where the etching process is performed by using a substrate processing apparatus provided with a cold-wall-type process furnace.

(102) Even in the case of using these substrate processing apparatuses, each processing may be performed with processing sequences and processing conditions similar to those of the embodiments described above to achieve effects similar to those of the aforementioned embodiments may be achieved.

(103) The embodiments described above may be appropriately combined with one another. The processing procedures and processing conditions at this time may be similar to, for example, the processing procedures and processing conditions of the aforementioned embodiments.

EXAMPLE

(104) A SiO film and a SiN film, which are bases exposed on a surface of a wafer, are etched and their etching amounts are measured respectively by using the aforementioned substrate processing apparatus and by the gas supply sequence illustrated in FIG. **4**. The processing condition is set to a predetermined condition which falls within a range of the processing condition described in the aforementioned embodiments.

(105) FIG. **6** shows measurement results of etching amounts. In FIG. **6**, a horizontal axis indicates the number of times that a cycle including steps 1 to 3 is performed and a vertical axis indicates an amount of change in a film thickness of the base, that is, an etching amount ($\text{\AA}$) of the base. In FIG. **6**, .square-solid. marks indicate a measurement result of the SiO film, and .diamond-solid. marks indicate a measurement result of the SiN film.

(106) As shown in FIG. **6**, it can be seen that, in both the SiO film and the SiN film, the etching amount per one cycle can be kept within the range of 0.5 to 1 $\text{\AA}$, that is, the etching amount of the base can be controlled at a level of, for example, one atomic layer (molecular layer) or less.

(107) According to the present disclosure in some embodiments, it is possible to enhance etching

controllability.

(108) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A processing method comprising: etching a base on a surface of a substrate by alternately performing: (a) forming a layer on a surface of the base by exposing the base to a modifying agent; and (b) causing a reaction between a halogen-containing radical and the base by exposing the layer to a halogen-containing gas such that the layer reacts with the halogen-containing gas to generate the halogen-containing radical.
2. The processing method according to claim 1, wherein in (b), the layer is exposed to the halogen-containing gas under a condition in which a reaction between the halogen-containing gas and the layer proceeds further than a reaction between the halogen-containing gas and the base.
3. The processing method according to claim 1, wherein in (b), the layer is exposed to the halogen-containing gas under a condition in which a reaction between the halogen-containing gas and the layer proceeds and a reaction between the halogen-containing gas and the base does not proceed.
4. The processing method according to claim 1, wherein in (b), the base is etched with the halogen-containing radical generated by a reaction between the halogen-containing gas and the layer.
5. The processing method according to claim 4, wherein in (b), the base is exposed to the halogen-containing radical under a condition in which the reaction between the halogen-containing radical and the base proceeds further than a reaction between the halogen-containing gas and the base.
6. The processing method according to claim 4, wherein in (b), the layer is exposed to the halogen-containing gas under a condition in which the reaction between the halogen-containing radical and the base proceeds and a reaction between the halogen-containing gas and the base does not proceed.
7. The processing method according to claim 1, wherein the modifying agent includes a silicon-containing gas, and wherein in (a), silicon contained in the silicon-containing gas is adsorbed on the surface of the base.
8. The processing method according to claim 1, wherein the modifying agent includes aminosilane, and wherein in (a), silicon contained in the aminosilane is adsorbed on the surface of the base.
9. The processing method according to claim 7, wherein in (a), the base is exposed to the modifying agent under a condition in which the adsorption of the silicon on the surface of the base is saturated.
10. The processing method according to claim 7, wherein in (a), the base is exposed to the modifying agent under a condition in which the adsorption of the silicon on the surface of the base is self-limited.
11. The processing method according to claim 8, wherein in (a), the base is exposed to the modifying agent under a condition in which the adsorption of the silicon on the surface of the base is saturated.
12. The processing method according to claim 1, wherein (c) pre-processing the surface of the base is performed before performing (a).
13. The processing method according to claim 12, wherein in (c), an adsorption site is formed on the surface of the base.
14. The processing method according to claim 1, wherein (c) OH-terminating the surface of the

base is performed before performing (a).

15. The processing method according to claim 12, wherein in (c), the substrate is exposed to a gas containing oxygen and hydrogen.

16. The processing method according to claim 1, wherein the halogen-containing gas includes at least one selected from the group of a fluorine-containing gas, a chlorine-containing gas, and an iodine-containing gas.

17. The processing method according to claim 1, wherein the halogen-containing radical includes at least one selected from the group of a fluorine-containing radical, a chlorine-containing radical, and an iodine-containing radical.

18. The processing method according to claim 1, wherein the base contains at least one selected from the group of a semi-metal element and a metal element.

19. The processing method according to claim 1, wherein the layer is a modified layer.

20. The processing method according to claim 1, wherein a cycle including (a) and (b) is performed a predetermined number of times in the etching.

21. The processing method according to claim 12, wherein a cycle including (a), (b) and (c) is performed a predetermined number of times in the etching.

22. A method of manufacturing a semiconductor device, comprising the processing method of claim 1.

23. A non-transitory computer-readable recording medium storing a program that causes, by a computer, a processing apparatus to perform a process comprising: etching a base on a surface of a substrate by alternately performing: (a) forming a layer on a surface of the base by exposing the base to a modifying agent; and (b) causing a reaction between a halogen-containing radical and the base by exposing the layer to a halogen-containing gas such that the layer reacts with the halogen-containing gas to generate the halogen-containing radical.
